



QUESTIONING
REALITY

UNIVERSITY OF
TORONTO

DATA SCIENCES
INSTITUTE

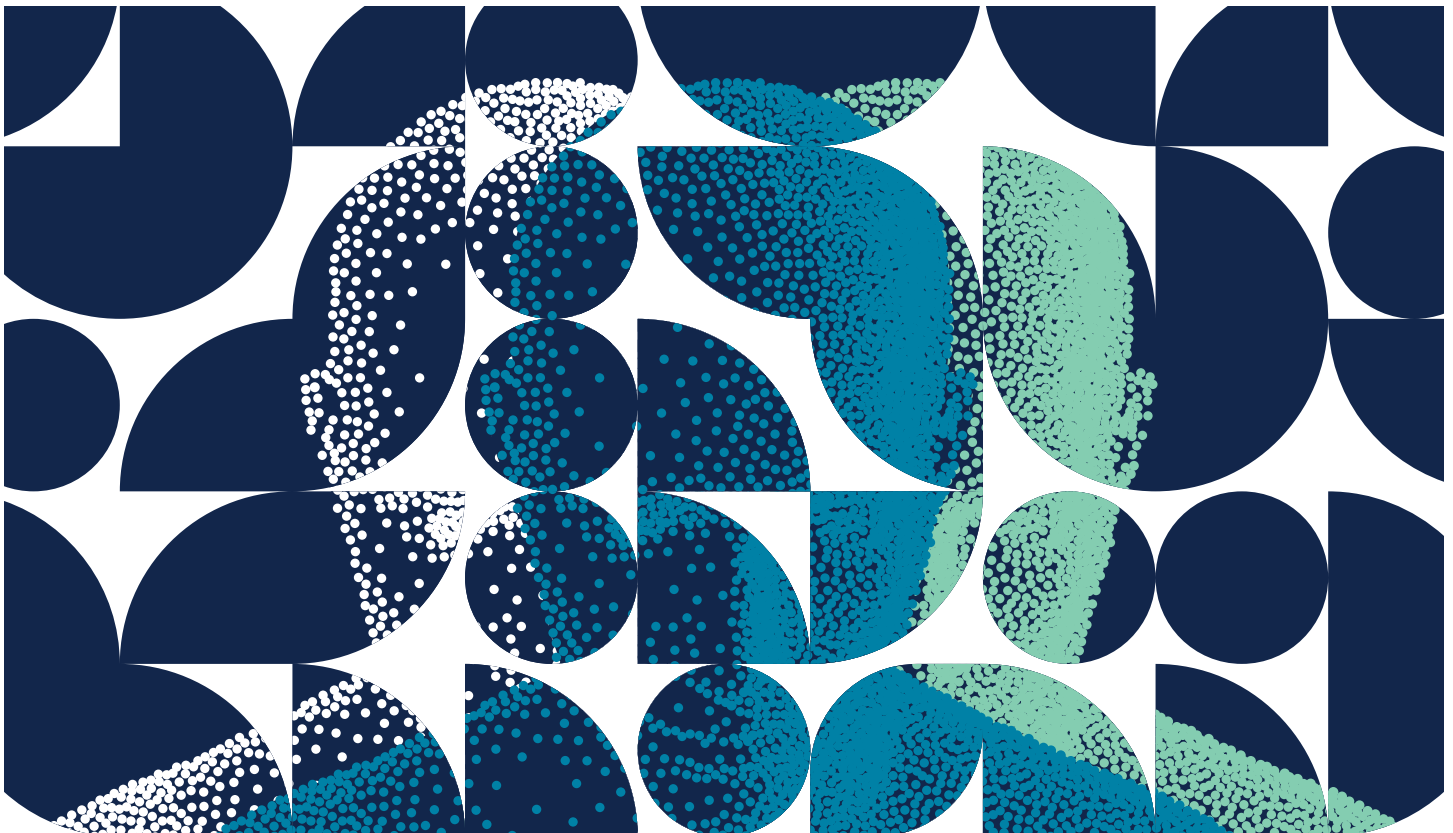


ALFRED P. SLOAN
FOUNDATION

WHITE PAPER

Designing Social VR in the Age of Generative AI

Mia Wang, University of Colorado Boulder
Maggie Liao, University of Georgia
Matthew J. A. Craig, Central Michigan University
Marco Dehnert, University of Arkansas
Josh Baldwin, Boston University
Sam Hartling, University of Toronto
Grace Ahn, University of Georgia
Bree McEwan, University of Toronto Mississauga



EXECUTIVE SUMMARY

Generative AI (GenAI) is rapidly reshaping the landscape of virtual reality (VR), moving immersive environments beyond static, pre-designed spaces toward dynamic, adaptive, and socially responsive systems. As advances in large language models, generative media, and AI-driven interaction accelerate, VR is increasingly positioned not only as a medium for experience but as a platform for intelligent, co-constructed environments between people and machines. This shift introduces new possibilities for how users create, navigate, and interact within virtual spaces, while also raising fundamental questions about governance, design, and human experience.

At the same time, the integration of GenAI into VR reflects could transform how virtual environments are produced and inhabited. For instance, GenAI can potentially lower the barriers to content creation, enabling users with limited technical expertise to generate immersive worlds, avatars, and interactions. From a communication perspective, GenAI has the potential for enhancing social and communicative affordances through real-time translation, adaptive conversational agents, and AI-driven personalization. These developments expand access to VR across social, physical, and economic dimensions, while enabling new forms of interaction that blur the boundaries between human-to-human and human-AI communication.

Despite these opportunities, significant uncertainty remains regarding how GenAI-enabled VR systems should be designed, implemented, and governed. Unlike earlier digital platforms, AI-driven VR environments continuously sense, interpret, and respond to user behavior, making them deeply embedded socio-technical systems. This raises interconnected challenges related to algorithmic biases and the potential homogenization of virtual experiences, heightened privacy risks from continuous behavioral and biometric data collection, and unresolved questions about intellectual property and ownership in generative systems. In addition, the integration of GenAI into VR production pipelines is reshaping creative labor, altering how work is distributed, valued, and controlled.

From an industry and policy perspective, these developments introduce complex decisions about adoption, oversight, and responsible innovation. Stakeholders must evaluate when and how GenAI meaningfully enhances VR experiences, how to balance personalization with user autonomy and privacy, and how to ensure that immersive systems remain inclusive, diverse, and socially legitimate. At the same time, there is a need for clearer frameworks to guide the role of AI within virtual environments—particularly in distinguishing when AI should augment human interaction versus when it risks replacing it.

This white paper addresses these challenges by synthesizing insights from the *2025 Questioning Reality* conference, where scholars, industry leaders, and technologists engaged in early-stage interdisciplinary discussions on the future of VR. Rather than offering prescriptive solutions, the paper identifies key opportunities, tensions, and design considerations shaping social VR. By articulating these themes, the white paper aims to inform future research and industry innovation agendas and support more intentional, ethical, and human-centered approaches to the integration of generative AI in virtual reality.

BACKGROUND ON THE QUESTIONING REALITY CONFERENCE

The Questioning Reality: Explorations of Virtual Reality conference was an in-person conference held at the Data Sciences Institute at the University of Toronto in April 2025. Co-organized by Dr. Sun Joo (Grace) Ahn, Director of the Center for Advanced Computer-Human Ecosystems (CACHE) and Professor of Advertising and Public Relations at the University of Georgia, USA, and Dr. Bree McEwan, Associate Director of the Data Sciences Institute and Associate Professor at the Institute for Communication, Culture, and Information Technology at the University of Toronto Mississauga, Canada. The conference was made possible through the generous support of the Alfred P. Sloan Foundation, a not-for-profit, mission-driven grantmaking institution dedicated to improving the welfare of all through the advancement of scientific knowledge.

Started in 2024, the invite-only QR conference brought together industry professionals, leading scholars, and enthusiasts working in the fields of virtual reality, augmented reality, extended reality, and mixed realities. Blair MacIntyre, Global Head of Immersive Technology Research at JP Morgan Chase, and Pablo Pérez, Lead Scientist at the Extended Reality Lab at Nokia, delivered stimulating keynotes. Rather than a typical academic conference where speakers present work that is already completed, QR aimed to have attendees come together at the beginning of the research process—to focus on spirited discussions, brainstorming and Blue Sky discussions, networking, and collaborative worldmaking. Core topics of the conference included generative artificial intelligence and VR as well as collaborative work and VR. The results of discussions around the prior are the focal point of this whitepaper. Specifically, this white paper synthesizes insights regarding the opportunities, challenges, and future directions that GenAI introduces to VR environments.

OPPORTUNITIES

Expanding Access: Alleviating Social & Physical Barriers

Generative AI has the potential to expand access to VR and democratize its use for a broader population. There are numerous barriers to VR adoption, including both physical and social constraints. Researchers have long examined ways to mitigate physical barriers such as motion sickness and discomfort associated with prolonged VR use, as well as hardware limitations and usability challenges (LaViola, 2000; Rebenitsch & Owen, 2016). At the same time, VR environments can be inherently isolating, as users may be unwilling to engage for extended periods when meaningful social interaction is lacking (e.g., difficulty finding others outside of centralized hub environments).

GenAI introduces new opportunities to simulate and enhance social interactions within virtual spaces. For instance, it enables creators to design large-scale, socially engaging events (e.g., virtual “social movie nights”), thereby facilitating interaction in a more efficient and accessible

manner. Through algorithmic personalization, AI can help users explore relevant communities, connect with like-minded individuals, and engage in more meaningful conversations. In addition, GenAI can support real-time social interactions by reducing communication barriers. One of the most prominent examples is language translation (e.g., “jokes being told in another language”): GenAI-powered translation systems can enable seamless cross-lingual communication, addressing one of the most significant obstacles to intercultural social interaction (Deardorff, 2006).

This capability may be particularly beneficial for individuals with high communication anxiety, lower communication competence, or neurodiverse conditions. GenAI systems can provide adaptive suggestions for how to respond in social contexts, thereby supporting users’ communicative efforts (e.g., “recommending how to approach social cues for someone on the autism spectrum; helping to read social cues”). In this sense, AI not only enhances VR-mediated interactions but can also augment real-world communication through Social XR or AR applications (“AI in AR will remove these language barriers”). By reducing social barriers, GenAI empowers users with communication constraints across both virtual and physical environments.

Enhancing Interaction: Human-AI Dynamics

In addition to facilitating human-human interaction, attendees identified significant opportunities for enhancing human-AI social interaction in VR, including functions such as “chatting about whatever with a friend,” “giving social advice,” “acting like a professional teaching you a skill set that you want to learn,” and serving as “a companion rather than an avatar on your behalf.” GenAI enables the proliferation of AI-powered avatars that can communicate and interact dynamically with users. These avatars can serve multiple roles depending on user needs. For example, for novice users unfamiliar with VR environments, a GenAI-powered avatar can function as a virtual tour guide, assisting with navigation and answering user queries in real time. In educational contexts, AI avatars can act as instructors, demonstrating skills and enabling interactive, conversational learning experiences.

Moreover, voice-based interaction adds an additional modality that enhances immersion and usability. For instance, in VR applications such as VTOL VR, users can issue voice commands to



© Margo Alexa - stock.adobe.com

interact with the environment (e.g., requesting assistance, coordinating actions), illustrating how conversational interfaces can deepen user engagement (Bowman & McMahan, 2007). Beyond functional roles, AI avatars may also provide social and emotional support. Increasing evidence suggests that users are turning to large language model (LLM)-based chatbots for mental health support, with controlled studies indicating that AI therapy can achieve outcomes comparable to those of human therapists (Heinz et al., 2025; Raile, 2024).

These effects may be amplified in VR contexts, where embodiment enhances the perceived presence of AI agents. Embodied avatars can increase social presence and facilitate stronger relational bonds, thereby improving emotional support and therapeutic engagement (Biocca et al., 2003; Oh et al., 2018). However, attendees also raised concerns about the potential for over-reliance on AI avatars, which may reduce meaningful human-human interactions (“it could be a replacement for all human interaction”). Accordingly, it is important to position GenAI-powered agents as facilitators that augment, rather than replace, interpersonal social interaction.

Improving Accessibility & Lowering Technical Barriers

Beyond social interaction, attendees discussed how GenAI can alleviate physical and accessibility-related barriers. Participants emphasized that “people with disabilities, for example, who can’t travel or participate in things they might want to participate in” could benefit integrating GenAI and VR, as these technologies “would allow people to participate” in otherwise inaccessible activities. As one example highlighted, users who “want to go skydiving but are unable to do so (too old, disabled, or because it is too expensive) can instead experience it through VR,” which “makes it more accessible because it is cheaper and does not have the same physical barriers.”

AI systems can dynamically adapt environments based on individual user needs, such as adjusting text size, brightness, or interface complexity. This allows for highly personalized experiences tailored to diverse accessibility requirements. Furthermore, GenAI significantly reduces the technical barriers to creating VR environments. Traditionally, developing VR content requires substantial time and expertise. However, AI-driven tools for generating 2D and 3D assets (e.g., text-to-3D models such as Genie 3) can streamline this process, enabling even novice users to create virtual worlds without extensive coding knowledge (Poole et al., 2022). Despite these advancements, attendees noted that current technologies remain limited in accuracy and contextual understanding. While GenAI can generate environments and assets, it may not yet achieve the desired level of precision or realism. This highlights the need for continued development of real-time environment generation and context-aware modeling to enhance social VR experiences further.

Expanding Research & Experiential Possibilities

In addition to improving user experiences, GenAI also has the potential to accelerate VR research. For researchers with limited programming expertise, particularly in fields such as social science, GenAI can facilitate the rapid creation of experimental stimuli, including virtual environments and avatars. However, attendees also highlighted challenges in validating AI-generated outputs. Novice researchers may struggle to identify implicit assumptions or errors embedded in generated code (e.g., “Genai increases efficiency and generates ‘what seems to be’ a viable

output”), which could affect the validity of experimental designs. Therefore, improving AI literacy and developing standardized validation practices are essential for responsible use.

Attendees further highlighted that GenAI enhances accessibility by expanding the range of VR experiences. VR already enables users to access experiences that may be physically or economically inaccessible in real life (e.g., extreme sports, travel). GenAI can further enhance this capability by enabling highly personalized simulations. For example, users who are unable to visit their hometowns can recreate personalized virtual environments using AI, such as applications integrated with geographic data (e.g., Google Earth-based platforms).

The benefits of GenAI are particularly pronounced for users with accessibility needs. For instance, social VR heavily relies on verbal communication, which may limit accessibility for some users. GenAI enables alternative modes of expression, such as visual storytelling and artistic representation, thereby introducing additional layers of communication. This multimodal approach allows users to convey ideas that may be difficult to express verbally.

Recent empirical research also demonstrates that GenAI can improve accessibility for blind and low-vision (BLV) users. Traditional VR accessibility solutions often rely on spatial audio or haptic feedback (Dang et al., 2023; Siu et al., 2020), which can be cognitively demanding and potentially disruptive in social contexts. In contrast, Collins et al. (2026) developed an LLM-powered virtual guide that assists BLV users in navigating VR environments. Their findings suggest that such AI guides are often perceived as social companions, encouraging interaction and engagement, and demonstrating the potential of GenAI avatars as versatile tools for accessibility.

Summary

In summary, attendees emphasized that Generative AI offers substantial potential to alleviate both social and physical barriers in VR. By enhancing accessibility, enabling personalization, and facilitating both human-human and human-AI interactions, GenAI significantly expands the possibilities of social VR environments. At the same time, these opportunities must be balanced with careful consideration of technical limitations, ethical concerns, and potential risks associated with over-reliance on AI systems. The next section will address these challenges in greater detail.

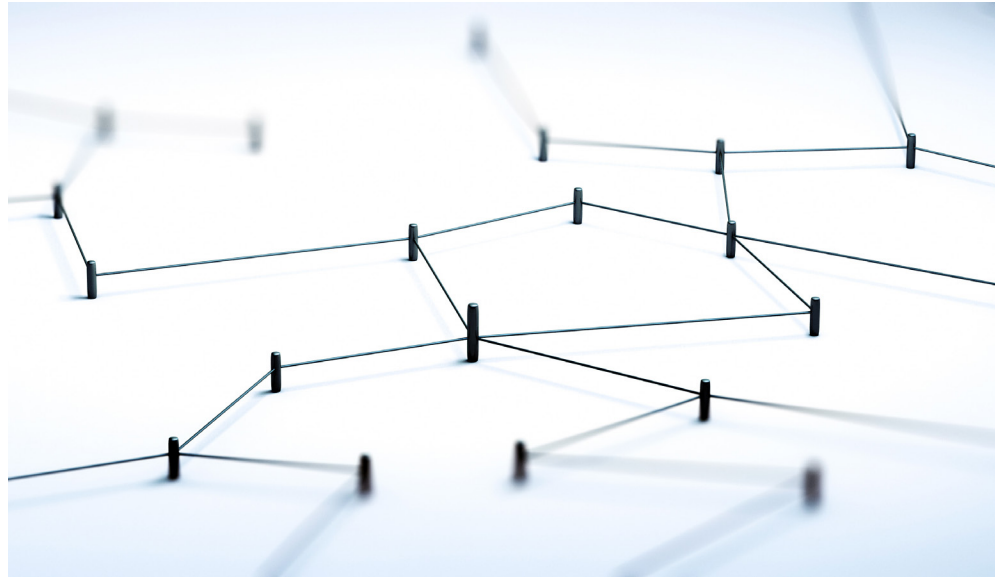
CHALLENGES

AI is rapidly transforming VR by enabling adaptive environments, intelligent agents, and scalable content production. However, the central challenge is not simply technological capability, but whether AI-enabled VR systems can remain socially legitimate, economically viable, and ethically governed when they continuously sense user behavior and shape immersive experiences in real time. In VR, technical constraints such as latency directly affect user comfort, while data collection becomes embodied behavioral surveillance, and algorithmic decisions influence not only content but perception itself (Radianti et al., 2020; Slater & Sanchez-Vives, 2016). This aligns with broader observations that AI in VR must be understood as a tightly coupled socio-technical system rather than a standalone innovation.

The challenges discussed below are interconnected and mutually reinforcing, spanning bias, privacy, intellectual property, labor, and psychosocial risks. Together, they shape the conditions under which AI can be responsibly integrated into immersive environments.

Bias and the Crisis of “Average”

AI models are often trained on datasets derived from Western, Educated, Industrialized, Rich, and Democratic (WEIRD) populations, leading to outputs that may not generalize globally (Henrich et al., 2010; Bender et al., 2021). This concern was clearly articulated in the discussion: “The AI bias might represent WEIRD... cultures... The mean will eliminate diversity.” In VR, such bias becomes more consequential



© 1xpert - stock.adobe.com

because it is embedded in spatial and embodied experiences, influencing avatars, environments, and interaction norms rather than simply textual outputs.

The lack of diversity among creators can potentially worsen the existing biases in VR. As noted, “People creating these environments are a small subset of the population (high educated white males).” This structural imbalance shapes what is considered “normal” in immersive environments, reinforcing dominant cultural assumptions. Evidence suggests that the broader technology and game development workforce—closely tied to VR production—remains demographically skewed, which has downstream effects on design and representation (U.S. Equal Employment Opportunity Commission, 2024).

A second, closely related challenge is the “crisis of average.” Generative AI systems optimize for probabilistic likelihood, producing outputs that reflect statistically common patterns. As described in the discussion, “Because of the averageness, it will wipe out all creativity... Crisis of bland generative content.” Empirical findings support this concern: while AI can enhance individual creative performance, it reduces diversity across outputs, leading to homogenization (Doshi & Hauser, 2024; Wenger & Kenett, 2026).

In VR, this risks “average worldbuilding,” where environments are technically polished but culturally flattened and aesthetically interchangeable. As immersive systems increasingly rely on generative pipelines, the loss of diversity may not manifest as overt bias, but as subtle standardization—where statistically dominant norms gradually replace difference. This suggests that the core issue is not only bias, but the structural tendency of AI to converge toward the mean.

Privacy Intrusiveness

Privacy risks in AI-enabled VR are significantly more invasive than in traditional digital systems due to the continuous capture of behavioral and biometric data. VR systems collect head movement, gaze, gestures, and voice, enabling AI to infer user states, preferences, and even latent traits. As noted in the discussion, “Too much friction they just ignore (skip terms and conditions)... certain threshold of information they just skip.” This reflects the phenomenon of privacy fatigue, where users disengage from privacy decision-making due to cognitive overload (Choi et al., 2018; c.f. Craig, 2025).

In VR, this problem is intensified because data collection is often implicit and unavoidable. Immersive systems rely on continuous sensing to function, meaning that users cannot easily opt out without losing access to the experience. As highlighted in recent analysis, behavioral data in VR is not merely supplementary—it is foundational to interaction. A key concern is behavioral fingerprinting. As the discussion notes, “Analyze zoom recordings of keyboard sound... can figure out what people are typing... can do that in real time.” While this example comes from a different context, it illustrates a broader principle: seemingly indirect signals can reveal highly sensitive information. Research shows that VR motion data alone can uniquely identify users and infer personal attributes with high accuracy (Miller et al., 2020; Nair et al., 2024).

The discussion also highlights predictive inference: “Story: Didn’t know she was pregnant yet, algorithm already knew... AI keeps learning user behavior.” This reflects a well-documented phenomenon in data analytics, where algorithms infer sensitive conditions before users are aware of them (Duhigg, 2012). In VR, the richness and continuity of data amplify this capability, making predictive inference both more powerful and more intrusive. These dynamics suggest that traditional consent-based privacy frameworks are insufficient. Instead, privacy in AI-enabled VR must be addressed at the architectural level, including data minimization, on-device processing, and restrictions on inferential use of behavioral data.

Intellectual Property Risks

The integration of generative AI into VR introduces significant challenges related to copyright and ownership. Current legal frameworks do not grant copyright protection to works that AI entirely generates without meaningful human authorship (U.S. Copyright Office, 2025). This creates uncertainty for creators and organizations seeking to use AI-generated content in commercial VR environments. As noted in the discussion, “No commercial incentive for generative AI environments because you cannot copyright anything.” While this statement overgeneralizes the legal reality, it captures an important concern: unclear ownership reduces investment incentives in high-quality immersive



© Garun Studios - stock.adobe.com

content. In practice, copyright may still apply to human-authored elements such as selection, arrangement, and modification, but the boundary between human and machine authorship remains ambiguous.

This uncertainty has broader implications for VR production pipelines. As highlighted in the integrated material, AI simultaneously lowers production costs while destabilizing ownership and value structures, creating tension between efficiency and appropriability. This dual effect complicates commercialization and may slow adoption in high-stakes applications.

Job Displacement

The adoption of AI in VR production also raises concerns about labor displacement, particularly in creative roles such as 3D modeling and environment design. As noted in the discussion, “Gen AI might be the same way—a disruptive tech that will replace a number of jobs eventually.”

Current evidence suggests that AI is more likely to restructure jobs rather than eliminate them entirely. However, it can significantly reduce demand for routine and entry-level creative tasks while shifting value toward higher-level conceptual and supervisory roles (Brynjolfsson et al., 2018). Industry data further indicates growing concern among creative professionals about the impact of generative AI on their work.

In the VR context, this transformation may lead to a concentration of creative control among those who design and manage AI systems, raising questions about equity, access, and the future of creative labor. The challenge is therefore not only economic but also structural, affecting how creativity is distributed and valued.

Physical & Social Isolation

AI-enabled VR systems also introduce significant psychosocial challenges, particularly related to isolation and human relationships. While immersive environments can facilitate connection, AI-driven personalization may reinforce avoidance behaviors and reduce engagement with the physical world. The discussion highlights this concern: “The AI might encourage you to adopt more dysfunctional coping strategies because of the way it is trained on you.” For individuals with social anxiety, AI systems that adapt to user preferences may remove challenging interactions, preventing personal growth. Research suggests that problematic use of social VR is associated with increased isolation and weaker real-world social support (Yao et al., 2025).

The discussion also raises broader concerns about AI relationships: “Will robots displace human reproduction? Safety in AI relationships.” While speculative, this reflects growing concern about emotional attachment to AI systems. Studies indicate that AI companions can provide short-term relief from loneliness but may also lead to emotional dependence and reduced well-being over time (Phang et al., 2025). In immersive environments, these effects may be amplified due to increased realism and presence. As noted in the integrated material, AI systems that optimize for engagement may prioritize comfort and validation, potentially weakening users’ ability to navigate real-world relationships.

Generative AI in VR introduces key socio-technical challenges, including bias and content homogenization, heightened privacy risks from continuous behavioral data collection, and unclear intellectual property ownership. It also reshapes creative labor while raising psychosocial concerns such as over-reliance on AI and potential social isolation, underscoring the need for careful governance beyond technical innovation.

IMPLICATIONS & FUTURE DIRECTIONS OF AI IN VR

The discussions at the Questioning Reality 2025 conference highlight that the future of Artificial Intelligence in Virtual Reality will be shaped not only by technological advancement, but by how these systems are designed, governed, and integrated into human environments. Three key directions emerge for guiding the responsible and effective development of AI in VR: preserving diversity in generative systems, defining the role of AI in social interaction, and preparing for AI-augmented creative labor.

Designing for Diversity in AI-Generated Worlds

As generative systems become central to VR content creation, a key implication is the need to actively design for diversity rather than relying on default model outputs. While AI lowers technical barriers and enables broader participation through practices such as “vibe coding,” it also introduces the risk of convergence toward standardized, “average” outputs.

This duality suggests that future VR systems should not only prioritize efficiency and accessibility but also incorporate mechanisms that preserve variation, cultural specificity, and creative originality. This may include diversifying training data, enabling user-driven customization, and developing tools that support exploration beyond algorithmically dominant patterns. In this context, the goal is not to resist automation, but to ensure that generative pipelines expand—rather than narrow—the range of possible virtual experiences.

Artificial Intelligence ‘Reading Glasses’

As AI avatars and conversational agents become increasingly embedded in VR environments, a central implication is the need to define their role in relation to human interaction. AI systems have the potential to function as social facilitators—supporting communication, providing guidance, and enhancing accessibility. However, without careful design, these same systems may also encourage over-reliance and reduce engagement with human relationships.



© DaliCeMedia - stock.adobe.com

This highlights the importance of designing AI as an augmentative layer rather than a substitute for human connection. The “reading glasses” analogy discussed by attendees illustrates this

balance: AI should assist users when needed, but not become a default or unavoidable intermediary in all social interactions. Future design strategies should therefore distinguish between contexts where AI enhances social capacity and those where it risks replacing meaningful human engagement.

Preparing for AI-Augmented Creative Labor

The integration of generative AI into VR production has significant implications for how creative work is structured and valued. While AI tools democratize content creation by enabling non-experts to generate environments and assets, they also reshape labor dynamics by reducing demand for routine production tasks and shifting emphasis toward higher-level conceptual and supervisory roles.

This transformation calls for a redefinition of creative workflows, where human expertise is increasingly focused on direction, curation, and critical evaluation rather than manual production. At the same time, it raises important questions about maintaining diversity of thought and preserving the human element in creative outputs. Future efforts should therefore focus on developing skill sets, educational models, and industry practices that support collaboration between human creators and AI systems, rather than simple substitution.

Conclusion

The future of artificial intelligence in virtual reality is not set in stone. Through the collaboration of computer scientists, developers, policy makers, social scientists, and users of this technology, there is a future for artificial intelligence in virtual reality that serves human needs and desires. The pending crisis of average, AI as reading glasses, and the labor of AI serve as stark reminders that complex technology integrating into even more complex human systems will require a level of human determination and creative problem-solving that is far beyond average.

REFERENCES

- Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021, March). On the dangers of stochastic parrots: Can language models be too big? 🦜 In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency* (pp. 610-623). <https://doi.org/10.1145/3442188.3445922>
- Biocca, F., Harms, C., & Burgoon, J. K. (2003). Toward a more robust theory and measure of social presence: Review and suggested Criteria. *Presence: Teleoperators and Virtual Environments*, 12(5), 456–480. <https://doi.org/10.1162/105474603322761270>
- Bowman, D. A., & McMahan, R. P. (2007). Virtual reality: How much immersion is enough? *Computer*, 40(7), 36–43. <https://doi.org/10.1109/MC.2007.257>
- Brynjolfsson, E., Mitchell, T., & Rock, D. (2018, May). What can machines learn and what does it mean for occupations and the economy? In *AEA papers and proceedings* (Vol. 108, pp. 43-47). <https://doi.org/10.1257/pandp.20181019>

Choi, H., Park, J., & Jung, Y. (2018). The role of privacy fatigue in online privacy behavior. *Computers in Human Behavior*, 81, 42–51. <https://doi.org/10.1016/j.chb.2017.12.001>

Collins, J., Lin, S. Y., Liu, T., Won, A. S., & Azenkot, S. (2026). Understanding the use of a large language model-powered guide to make virtual reality accessible for blind and low vision people. *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems, CHI '26*, 1–16. <https://doi.org/10.1145/3772318.3791143>

Craig, M. J. (2025). Human-machine communication privacy management, privacy fatigue, and the conditional effects of algorithm awareness on privacy co-ownership in the social media context. *Computers in Human Behavior*, 173, 108786. <https://doi.org/10.1016/j.chb.2025.108786>

Dang, K., Korreshi, H., Iqbal, Y., & Lee, S. (2023). Opportunities for accessible virtual reality design for immersive musical performances for blind and low-vision people. *Proceedings of the 2023 ACM Symposium on Spatial User Interaction, SUI '23*, 1–21. <https://doi.org/10.1145/3607822.3614540>

Deardorff, D. K. (2006). Identification and assessment of intercultural competence as a student outcome of internationalization. *Journal of Studies in International Education*, 10(3), 241–266. <https://doi.org/10.1177/1028315306287002>

Doshi, A. R., & Hauser, O. P. (2024). Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science advances*, 10(28), eadn5290. <https://doi.org/10.1126/sciadv.adn5290>

Duhigg, C. (2012). *The power of habit: Why we do what we do in life and business* (Vol. 34, No. 10). Random House.

Heinz, M. V., Mackin, D. M., Trudeau, B. M., Bhattacharya, S., Wang, Y., Banta, H. A., Jewett, A. D., Salzhauser, A. J., Griffin, T. Z., & Jacobson, N. C. (2025). Randomized trial of a generative AI chatbot for mental health treatment. *NEJM AI*, 2(4), Aloa2400802. <https://doi.org/10.1056/Aloa2400802>

Henrich, J., Heine, S. J., & Norenzayan, A. (2010). The weirdest people in the world? *Behavioral and brain sciences*, 33(2-3), 61–83. <https://doi.org/10.1017/S0140525X0999152X>

LaViola, J. J. (2000). A discussion of cybersickness in virtual environments. *SIGCHI Bull.*, 32(1), 47–56. <https://doi.org/10.1145/333329.333344>

Miller, M. R., Herrera, F., Jun, H., Landay, J. A., & Bailenson, J. N. (2020). Personal identifiability of user tracking data during observation of 360-degree VR video. *Scientific Reports*, 10(1), 17404. <https://doi.org/10.1038/s41598-020-74486-y>

Nair, V., Rack, C., Guo, W., Wang, R., Li, S., Huang, B., ... & Song, D. (2024, March). Inferring private personal attributes of virtual reality users from ecologically valid head and hand motion data. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 477–484). IEEE. <https://doi.org/10.1109/VRW62533.2024.00094>

Oh, C. S., Bailenson, J. N., & Welch, G. F. (2018). A systematic review of social presence: Definition, antecedents, and implications. *Frontiers in Robotics and AI*, 5. <https://doi.org/10.3389/frobt.2018.00114>

Phang, J., Lampe, M., Ahmad, L., Agarwal, S., Fang, C. M., Liu, A. R., Danry, V., Lee, E., Pataranutaporn, P., & Maes, P. (2025). Investigating affective use and emotional well-being on ChatGPT. *OpenAI & MIT Media Lab*. <https://doi.org/10.48550/arXiv.2504.03888>

Poole, B., Jain, A., Barron, J. T., & Mildenhall, B. (2022). DreamFusion: Text-to-3D using 2D Diffusion (arXiv:2209.14988). *arXiv*. <https://doi.org/10.48550/arXiv.2209.14988>

Radianti, J., Majchrzak, T. A., Fromm, J., & Wohlgenannt, I. (2020). A systematic review of immersive virtual reality applications for higher education: Design elements, lessons learned, and research agenda. *Computers & Education*, 147, 103778. <https://doi.org/10.1016/j.compedu.2019.103778>

Raile, P. (2024). The usefulness of ChatGPT for psychotherapists and patients. *Humanities and Social Sciences Communications*, 11(1), 47. <https://doi.org/10.1057/s41599-023-02567-0>

Rebenitsch, L., & Owen, C. (2016). Review on cybersickness in applications and visual displays. *Virtual Reality*, 20(2), 101–125. <https://doi.org/10.1007/s10055-016-0285-9>

Siu, A. F., Sinclair, M., Kovacs, R., Ofek, E., Holz, C., & Cutrell, E. (2020). Virtual reality without vision: A haptic and auditory white cane to navigate complex virtual worlds. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems, CHI '20*, 1–13. <https://doi.org/10.1145/3313831.3376353>

Slater, M., & Sanchez-Vives, M. V. (2016). Enhancing our lives with immersive virtual reality. *Frontiers in Robotics and AI*, 3, 74. <https://doi.org/10.3389/frobt.2016.00074>

U.S. Copyright Office. (2025). *Copyright and artificial intelligence: Part 2—Copyrightability*. <https://www.copyright.gov>

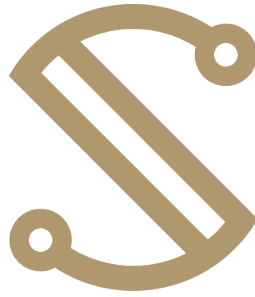
U.S. Equal Employment Opportunity Commission. (2024). *U.S. Equal Employment Opportunity Commission. (2024). High tech, low inclusion: Diversity in the high tech workforce and sector (2014–2022)*. <https://www.eeoc.gov>

Wenger, E., & Kenett, Y. N. (2026). Large language models are homogeneously creative. *PNAS nexus*, 5(3), pgag042. <https://doi.org/10.1093/pnasnexus/pgag042>

Yao, S. X., Lee, J., Reynolds, R. M., & Ellithorpe, M. E. (2025). Problematic social media use in 3D? Relationships between traditional social media use, social virtual reality (VR) use, and mental health. *PloS one*, 20(1), e0314863. <https://doi.org/10.1371/journal.pone.0314863>

CONTRIBUTORS

Farzan Rahimi, MacEwan University, Canada
Robby Ratan, Michigan State University, USA
Raheleh Saryazdi, Trent University, Canada
Shima Dadkhahfard, MacEwan University, Canada
Mia Wang, University of Colorado, USA
Michelle Lui, University of Toronto, Canada
Michael Nixon, University of Toronto, Canada
Gary Bader, University of Toronto, Canada
Sho Conte, University of Toronto, Canada
Adam Frost, University of Toronto, Canada
Eugene Kukshinov, University of Waterloo, Canada
Gigi Wong, Western University, Canada
Yoon Lee, Syracuse University, USA
Miguel Barreda-Angeles, European Commission, Italy
Mark Roman Miller, Illinois Institute of Technology, USA
Sue Lim, Michigan State University, USA
Tammy Lin, National ChengChi University, Taiwan
Diego Gomez-Zara, University of Notre Dame, USA
Mary Beth Oliver, Pennsylvania State University, USA
Eugy Han, Stanford, USA
Lindsay Hahn, SUNY-Buffalo, USA
Jesse Fox, The Ohio State University, USA
Katherine Isbister, UC-Santa Cruz, USA
Marco Denhart, University of Arkansas, USA
Joomi Lee, University of Arkansas, USA
Kristine Nowak, University of Connecticut, USA
Josh Baldwin, University of Georgia, USA
Kyle Johnsen, University of Georgia, USA
Tony Liao, University of Houston, USA
Christopher Ball, University of Illinois, USA
Breigha Adeymo, University of Illinois-Chicago, USA
Anna Queiroz, University of Miami, USA
Danny Pimental, University of Oregon, USA
Kuo-Ting Huang, University of Pittsburgh, USA
Matthew Craig, University of Tennessee, USA
Paul Alexander, University of Toronto, Canada
Luke Stark, Western University, Canada
Blake Markel, University of Toronto, Canada
Blair MacIntyre, JP Morgan Chase, USA
Pablo Perez, Nokia, Spain
Matthew Wren, VRAR Chicago, USA
Maxwell McGee, ENGAGE, USA
Jie Li, EPAM Systems, The Netherlands
Jamie She, Independent, USA
Karen Alexander, XR Women, USA



ALFRED P. SLOAN FOUNDATION

ACKNOWLEDGEMENT

Special thanks and appreciation for the Alfred P. Sloan Foundation for making this white-paper and the Questioning Reality Conference possible.